

## DS 411 : Optimization for Data Science

Course Code : DS 411

Course Name : Optimization for Data Science

L-T-P-C : 3-1-0-4

Intended for : Discipline core for B. Tech. Data Science and Engineering, Discipline elective for B. Tech. Computer Science and Engineering, Electrical Engineering, Free elective for other B. Tech. disciplines

Prerequisite : IC111-Linear Algebra, IC110-Engineering Mathematics, DS301- Mathematical Foundations of Data Science I

Mutual Exclusion:

Approval: 55th BoA

### Course Contents

- **Module I:** Affine sets, convex sets, cone, examples – hyperplanes, halfspaces, polyhedra, simplexes, positive semidefinite cones. Operations that preserve convexity. Separating and supporting hyperplanes. Dual cones. (6 lectures)
- **Module II:** Convex function, first and second order conditions, epigraph, operations that preserve convexity, conjugate function. (6 lectures)
- **Module III:** Convex optimization – linear, quadratic, geometric, conic, semidefinite programming. Formulation of - unconstrained, equality constrained, inequality constrained and both – problems. (7 lectures)
- **Module VI:** Duality – Lagrange dual function, bounds on the optimal value. Lagrange dual problem, weak and strong duality, optimality conditions. (8 lectures)
- **Module V:** Gradient methods – gradient descent, Lipschitz functions, smooth functions, projected gradient descent, Frank-Wolfe algorithm, Chebyhsev iterations, conjugate gradient, Nesterov's accelerated gradient descent. Dual gradient ascent, ADAM. (10 lectures)
- **Module VI:** Nonconvex optimization – alternating minimization and expectation maximization algorithms, convex relaxations. (5 lectures)

### Textbooks:

1. Boyd, Stephen, and Lieven Vandenberghe, **Convex optimization**, Cambridge university press, 2004.

### References:

1. Yurii, Nesterov, **Introductory lectures on convex optimization: a basic course**, Kluwer Academic Publishers, 2004.
2. Luenberger, D. G., and Y. Ye, **Linear and nonlinear programming**, Springer, 2008.
3. Nocedal, Jorge, and Stephen Wright, **Numerical optimization**, Springer Science & Business Media, 2006.